MAT2002 Ordinary Differential Equations
Final Review for slide01-15

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Definition of ODE and IVP

Definition 11.1

An ordinary differential equation is an equation involving **ONE** independent variable $t \in I$(I is an interval) and **ONE** dependent variable y of the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0.$$

Given constants $t_0, t_1, \ldots, t_{n-1} \in I$ and $y_0, y_1, \ldots, y_{n-1} \in \mathbb{R}$,we call

$$\begin{cases} F(t, y, y', y'', \ldots, y^{(n)}) = 0, \\ y(t_0) = y_0, \frac{dy}{dt}(t_1) = y_1, \ldots, \frac{d^{(n-1)}y}{dt^{n-1}}(t_{n-1}) = y_{n-1}, \end{cases}$$

an **initial value problem** (IVP).

The **order** of an ODE is the **highest order** of derivative in the ODE.

Definition related to ODE

Definition 11.2

(a) An ODE $F(t, y, y', y'', \ldots, y^{(n)}) = 0$ is **linear** is if F is a **linear function** of $y, \frac{dy}{dt}, \ldots, \frac{d^n y}{dt^n}$. Otherwise, it is a **non-linear** ODE. The general **linear** ODE of order n is

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)}(t) + \cdots + a_0(t)y(t) = f(t),$$

for some given functions $a_0, a_1, \ldots, a_n$ and f.

(b) An ODE is called **autonomous** if the independent variable does not appear explicitly (only in the derivatives), the **autonomous** ODE has the form: $F(y, y', y'', \ldots, y^{(n)}) = 0$. Otherwise it is a **non-autonomous** ODE.

1st-order linear ODE

$$\begin{cases} \frac{dy}{dt} = p(t)y + q(t), \\ y(t_0) = y_0, \end{cases} \tag{1}$$

General solution:

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)q(t)dt + c\right], \quad \text{where} \quad \mu(t) = \exp\left(-\int p(t)dt\right). \tag{2}$$

In order to use the initial condition more easily, the general solution can be rewritten as

$$y(t) = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(t)q(t)dt + c\right], \quad \text{where} \quad \mu(t) = \exp\left(-\int_{t_0}^{t} p(t)dt\right). \tag{3}$$

Using the initial condition, one can get $c = y_0$. The solution to IVP is

$$y(t) = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(t)q(t)dt + y_0\right], \quad \text{where} \quad \mu(t) = \exp\left(-\int_{t_0}^{t} p(t)dt\right). \tag{4}$$

Nonlinear to Linear: Bernoulli equation

Let n be a real number, $n \neq 0, 1$, and $p(t), q(t)$ be given functions. The **Bernoulli equation** is a first order non-linear ODE of the form

$$\boxed{\frac{dy}{dt} + p(t)y = q(t)y^n}. \tag{5}$$

As this is a non-linear ODE we cannot use integrating factors. Moreover, it doesn't seem like the equation is separable. Let us move the nonlinearity y^n to the derivative by multiplying the whole equation with y^{-n}:

$$y^{-n}\frac{dy}{dt} + p(t)y^{1-n} = q(t). \tag{6}$$

Now recognise that

$$\frac{d}{dt}\left(y^{1-n}\right) = (1-n)y^{-n}\frac{dy}{dt}.$$

Nonlinear to Linear: Bernoulli equation

So (6) can be simplified to

$$\frac{d}{dt}y^{1-n} + (1-n)p(t)y^{1-n} = (1-n)q(t). \tag{7}$$

Then, considering a **new variable** $v(t) = y^{1-n}(t)$, (7) becomes

$$\boxed{\frac{dv}{dt} + P(t)v = Q(t)}, \quad P(t) = (1-n)p(t), \quad Q(t) = (1-n)q(t), \tag{8}$$

which is a **linear** ODE for the variable v, and we can use integrating factors to solve. Let $\mu(t)$ be the integrating factor for (8), then the general solution is

$$v(t) = \frac{1}{\mu(t)}\left[\int Q(t)\mu(t)dt + c\right]$$

$$\Rightarrow \boxed{y(t) = \left(\frac{1}{\mu(t)}\left[\int Q(t)\mu(t)dt + c\right]\right)^{\frac{1}{1-n}}}.$$

1st-order non-linear ODE-separable equation

For 1st-order non-linear ODE, a general method is still missing. We can only solve some special type of 1st-order non-linear ODEs.

Definition 11.3

(Separable equation). A first order ODE $y' = f(t, y)$ is **separable** if it can be written in the form

$$M(t) + N(y)\frac{dy}{dt} = 0 \tag{9}$$

for some functions M and N.

Suppose there exist functions m and n such that

$$m' = M, \quad n' = N.$$

Then (9) can be written as

$$\frac{d}{dt}m(t) + \frac{d}{dt}n(y(t)) = 0.$$

Integrating yields the general (implicit) solution

$$\boxed{m(t) + n(y(t)) = c}, \quad c \in \mathbb{R} \tag{10}$$

Homogeneous equations

$$\frac{dy}{dt} = F(y/t).$$

Define a new variable $v = y/t \Leftrightarrow y = vt$. Then, the RHS of the ODE becomes just $F(v)$. For the LHS, by the product rule

$$y(t) = tv(t) \Rightarrow \frac{dy}{dt} = t\frac{dv}{dt} + v$$

$$\Rightarrow \boxed{t\frac{dv}{dt} + v(t) = F(v)}.$$

The transformed ODE in the variable v is now

$$\frac{dv}{dt} = \frac{F(v) - v}{t} \Rightarrow \boxed{\frac{1}{F(v) - v}\frac{dv}{dt} = \frac{1}{t}},$$

which is a **separable equation**!

1st-order non-linear ODE-Exact equation

Definition 11.4

(Exact equation). A first order ODE $M(t,y)+N(t,y)\frac{dy}{dt}=0$ is an **exact equation** if there exists a function $\Psi(t,y)$ such that

$$\boxed{\frac{\partial \Psi}{\partial t}(t,y)=M(t,y), \quad \frac{\partial \Psi}{\partial y}(t,y)=N(t,y)}. \tag{11}$$

The general solution $y(t)$ to the ODE is given implicitly as $\Psi(t,y(t))=c,\ c\in\mathbb{R}$.

Remark:
$M(t,y)+N(t,y)\frac{dy}{dt}=0$ (M_y and N_t are continuous) is exact $\Leftrightarrow M_y=N_t$.

1st-order non-linear ODE-(non-Exact equation)

If $M(t,y)+N(t,y)\frac{dy}{dt}=0$ is non-exact equation. Try to look for integrating factor $\mu(t,y)$ s.t. $\mu M(t,y)+\mu N(t,y)\frac{dy}{dt}=0$ is exact $\Leftrightarrow (\mu M)_y=(\mu N)_t$.

(1) $\mu(t,y)=\mu(t)$. Compute $K(t,y)=\frac{M_y-N_t}{N}(t,y)$;
If $K(t,y)=K(t)$, take $\mu(t)=e^{\int K(t)dt}$.

(2) $\mu(t,y)=\mu(y)$. Compute $H(t,y)=\frac{N_t-M_y}{M}(t,y)$;
If $H(t,y)=H(y)$, take $\mu(y)=e^{\int H(y)dy}$.

For 1st order nonlinear non-exact ODE, there is no general method.

Summary for ODEs we have learnt

Linear vs Nonlinear ODEs - a comparison

Type	Method	Explicit/Implicit solution
$y' = p(t)y + q(t)$	Integrating factor	$y(t) = \mu(t)^{-1}(\int \mu(t)q(t)dt + c)$
$M(t) + N(y)y' = 0$	Separable equation	$m(t) + n(y(t)) = c$
$y' + p(t)y = q(t)y^n$	$v := y^{1-n}$	$y(t) = (\mu^{-1}(\int Q(t)\mu(t)dt + c)^{1/(1-n)}$
$y' = F(y/t)$	$v = y/t$	$1/(F(v) - v)\frac{dv}{dt} = \frac{1}{t}$
$M(t, y) + N(t, y)y' = 0$	Exact equation	$\Psi(t, y(t)) = c$

Existence and Uniqueness

Next, we will address the theoretical fundamental question, is the 1st order nonlinear ODE $y' = f(t, y), y(t_0) = y_0$ has a solution? If it is, is the solution the only solution?

Existence and Uniqueness for first order linear ODE

Theorem 11.5

(Existence and Uniqueness for first order linear ODEs).
Suppose functions p and q are **continuous** *on $(\alpha, \beta) \subset \mathbb{R}$ (α, β are some real numbers). Then, for any $t_0 \in (\alpha, \beta), y_0 \in \mathbb{R}$, there* **exists** *a* **unique** *function $y(t)$ satisfying*

$$(*)\frac{dy}{dt} = p(t)y + q(t), \quad \forall t \in (\alpha, \beta),$$
$$y(t_0) = y_0.$$

And the solution is defined throughout the interval (α, β).

The solution **globally** exists in the interval (α, β) in which p and q are continuous.

Existence and uniqueness of the solution

Theorem 11.6 (Existence and uniqueness of the solution)

Consider the initial value problem (IVP)

$$\frac{dy}{dt} = f(t,y), \quad y(t_0) = y_0. - - - - - - - - - (*)$$

Let R be a closed rectangle

$$R = \{(t,y) || t - t_0| \leq a, \quad |y - y_0| \leq b\} (a > 0, b > 0).$$

(a, b are given positive constants)
Assume that $f(t,y)$ and $\frac{\partial f}{\partial y}$ are continuous on R.
Then the IVP has a unique solution $y = y(t)$ on the interval $[t_0 - h, t_0 + h]$ where $h = \min\left(\frac{b}{M}, a\right)$ and $M = \max\limits_{(t,y)\in R} |f(t,y)|$.

Existence and uniqueness of the solution

Integrating (*) from t_0 to t, one has
$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds. - - - - - - - - (**)$
Indeed, it can be easily shown that (*) and (**) are equivalent.
One method of showing that the integral equation (**) has a unique solution is known as the **Picard's iterative method**. First, we define the following successive iterations $\{\phi_n(t)\}_{n=0}^{\infty}$.

$$\phi_0(t) = y_0.$$

$$\phi_1(t) = y_0 + \int_{t_0}^{t} f(s, \phi_0(s))ds.$$

$$\vdots$$

$$\phi_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, \phi_n(s))ds$$

Non-uniqueness for first order non-linear ODEs

Remark: If $f(t, y)$ is continuous on the rectangle R containing (t_0, y_0). Then it can be shown that $\frac{dy}{dt} = f(t, y), y(t_0) = y_0$ has a solution, but uniqueness of the solution is not guaranteed.

Recall: Example (Non-uniqueness)

$$\frac{dy}{dt} = y^{1/3}, \quad y(0) = 0.$$

$f(t, y) = y^{1/3}$ is continuous, while $\frac{\partial f}{\partial y}(t, y) = \frac{1}{3} y^{-\frac{2}{3}}$ is not continuous at $y = 0$. Thus, the IVP has at least one solution, but the uniqueness is not guaranteed. Indeed, it has ∞ solutions.

Non-uniqueness for first order non-linear ODEs

Example (Non-uniqueness)

Since the ODE is separable, we obtain as a particular solution

$$y(t) = \left[\frac{2}{3}t\right]^{3/2} \quad \text{for } t \geq 0.$$

But note that the function $y_1(t) \equiv 0$ is also another solution, so is the function $y_2(t) = -\left[\frac{2}{3}t\right]^{3/2}$, and for arbitrary positive t_* the family of functions

$$y_*(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq t_*, \\ \pm\left[\frac{2}{3}(t - t_*)\right]^{3/2} & \text{if } t \geq t_*, \end{cases}$$

also solves the IVP. In particular we have found an **infinite** family of solutions to the IVP.

Second-order ODE

A second order ODE has the form

$$y'' = F(t, y, y')$$

If $F = g(t) - p(t)y' - q(t)y$, then the ODE is a linear ODE. Otherwise the ODE is nonlinear.
We mainly focus on the second-order linear ODE

$$y'' + p(t)y' + q(t)y = g(t).$$

Theorem 11.7

Consider the IVP

$$y'' + p(t)y' + q(t)y = r(t), \quad y(t_0) = y_0, \quad y'(t_0) = y_1.$$

Suppose $I = (\alpha, \beta) \subset \mathbb{R}$ is any open interval such that $t_0 \in I$, and the functions p, q, r are **continuous** *in I. Then, there is* **exactly one** *solution $y(t)$ to the IVP for $t \in I$.*

Homogeneous second-order ODE

Consider

$$y'' + p(t)y' + q(t)y = 0. \tag{12}$$

Definition 11.8

Given $y_1(t)$ and $y_2(t)$,

$$W[y_1, y_2](t) = \left| \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} \right|$$

is called the Wronskian for y_1 and y_2.

Homogeneous second-order ODE

Theorem 11.9

Suppose that I be an open interval in which $p(t)$ and $q(t)$ are continuous. Let $y_1(t)$ and $y_2(t)$ be two solutions to the ODE

$$y'' + p(t)y' + q(t)y = 0$$

for $t \in I$. Then, any solution $y(t)$ to the ODE can be expressed as

$$y(t) = c_1 y_1(t) + c_2 y_2(t) \tag{13}$$

for constants c_1 and $c_2 \Leftrightarrow \exists t_0 \in I$ such that the Wronskian $W(y_1, y_2)[t_0] \neq 0$. In this case, we say that (y_1, y_2) form a **fundamental set of solutions** *(FSS) to the ODE.*

Homogeneous second-order ODE

How to find the FSS for

$$ay'' + by' + cy = 0 \tag{14}$$

Look for solution form $y(t) = e^{rt}$

$$\boxed{ar^2 + br + c = 0}. \tag{15}$$

Case 1: Two distinct real roots $\Delta > 0$

In the case $b^2 > 4ac$, we obtain two real roots

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

This gives us two solutions that form a FSS

$$y_1(t) = \exp(r_1 t), \quad y_2(t) = \exp(r_2 t).$$

Case 2: Complex roots $\Delta < 0$

We now consider the case $\Delta = b^2 - 4ac < 0$. Then, the roots to the characteristic equation $ar_2 + br + c = 0$ is a **complex-conjugate pair**:

$$r_1 = \lambda + i\mu,\ \lambda = \frac{-b}{2a},\ \mu = \sqrt{4ac - b^2},\ i := \sqrt{-1},\ r_2 = \overline{r_1} = \lambda - i\mu.$$

Since the characteristic equation is obtained by substituting the trial function $y(t) = \exp(rt)$, we obtain two functions

$$y_1(t) = \exp(r_1 t) = \exp((\lambda + i\mu)t), \quad y_2(t) = \exp(r_2 t) = \exp((\lambda - i\mu)t).$$

Real-valued solutions $z_1(t), z_2(t)$ form a FSS:

$$z_1(t) = \exp(\lambda t)cos(\mu t),\ z_2(t) = \exp(\lambda t)\sin(\mu t).$$

Case 3: One repeated real root $\Delta = 0$

The last case is when $b^2 - 4ac = 0$ and we have a **repeated** root to the characteristic equation. The quadratic formula yields

$$r_1 = r_2 = -\frac{b}{2a}$$

$$y_1(t) = \exp\left(-\frac{b}{2a}t\right), y_2(t) = te^{-\frac{b}{2a}t}$$

$y_1(t), y_2(t)$ form a FSS.

Non-homogeneous second-order equations

We now turn our attention to ODE of the form

$$y'' + p(t)y' + q(t)y = r(t), \tag{16}$$

for given functions p, q and r that are continuous in an interval I. The corresponding **homogeneous** equation is

$$y'' + p(t)y' + q(t)y = 0. \tag{17}$$

Let $Y(t)$ denote a solution to (16), then **any solution** y to (16) can be expressed as

$$y(t) = Y(t) + c_1 y_1(t) + c_2 y_2(t),$$

where (y_1, y_2) is a fundamental set of solutions to the homogeneous problem (17).

Summary-undetermined coefficients

For

$$ay'' + by' + cy = r(t)$$

the trial function $Y(t)$ vs. $r(t)$ is listed as follows:

$r(t)$	Y(t)	The value for s
$P_n(t)e^{\alpha t}$	$Q_n(t)t^s e^{\alpha t}$	$s = \begin{cases} 0, \alpha \text{ is not a root.} \\ 1, \alpha = r_1 \neq r_2 \\ 2, \alpha = r_1 = r_2 \end{cases}$ r_1, r_2 are roots of $ar^2 + br + c = 0$

$$\begin{cases} P_n e^{\alpha t} \sin \beta t \\ P_n e^{\alpha t} \cos \beta t \end{cases} \quad \begin{cases} [Q_n(t) \cos \beta t \\ + R_n(t) \sin \beta t] t^s e^{\alpha t} \end{cases} \quad s = \begin{cases} 0, \text{if } \alpha + i\beta \text{ is not a root of } ar^2 + br + c = 0. \\ 1, \text{if } \alpha + i\beta \text{ is a root of } ar^2 + br + c = 0. \end{cases}$$

Variation of parameter-second-order equations

We now outline a general theory. Consider a general 2nd-order linear ODE

$$y'' + p(t)y' + q(t)y = r(t), \tag{18}$$

and suppose (y_1, y_2) forms a fundamental set of solutions to the homogeneous equation

$$y'' + p(t)y' + q(t)y = 0.$$

But how to find a particular solution to the non-homogeneous equation (18)?

Variation of parameter-second-order equations

The idea is as follows. Consider for some functions $u_1(t), u_2(t)$ such that the new function

$$y(t) = u_1(t)y_1(t) + u_2(t)y_2(t) \tag{19}$$

solves the non-homogeneous equation (18).

Variation of parameter-second-order equations

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ r \end{pmatrix}$$

$$u_1'(t) = -\frac{y_2 r}{W(y_1, y_2)}(t), \quad u_2'(t) = \frac{y_1 r}{W(y_1, y_2)}(t). \tag{20}$$

Integrating gives

$$u_1(t) = -\int \frac{y_2 r}{W(y_1, y_2)}(t)dt + d_1, \quad u_2(t) = \int \frac{y_1 r}{W(y_1, y_2)}(t)dt + d_2, \tag{21}$$

for constants $d_1, d_2 \in \mathbb{R}$.

Variation of parameter-second-order equations

Let us summarise with a theorem.

Theorem 11.10

Let $I \subset \mathbb{R}$ be an open interval, p, q, r continuous on I. If (y_1, y_2) is a fundamental set of solutions to the homogeneous equation $y'' + p(t)y' + q(t)y = 0$, then a particular solution to the non-homogeneous equation $y'' + p(t)y' + q(t)y = r(t)$ is

$$Y(t) = -y_1 \int \frac{y_2 r}{W(y_1, y_2)}(t)dt + y_2 \int \frac{y_1 r}{W(y_1, y_2)}(t)dt,$$

and the general solution to the non-homogeneous equation is

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t)$$

for constants $c_1, c_2 \in \mathbb{R}$.

General theory

The general nth order linear ODE is of the form

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = g(t),$$

and for an IVP we provide initial conditions

$$y(t_0) = x_0, \quad y'(t_0) = x_1, \quad \ldots, \quad y^{(n-1)}(t_0) = x_{n-1}.$$

We first state the existence and uniqueness theorem.

Theorem 11.11

(Existence and Uniqueness.) *Let $I \subset \mathbb{R}$ be an open interval and suppose $g, p_0, p_1, \ldots, p_{n-1}$ are continuous functions in I. For $t_0 \in I$ and $x_0, \ldots, x_{n-1} \in \mathbb{R}$, there is exactly one solution to the IVP*

$$\begin{cases} y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = g(t), \\ y(t_0) = x_0, \quad y'(t_0) = x_1, \quad \ldots, \quad y^{(n-1)}(t_0) = x_{n-1}. \end{cases}$$

High-order linear ODE

The general nth order linear ODE is of the form

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = g(t),$$

Definition 11.12

Given functions $f_1, \ldots, f_n$ that are differentiable up to order $n-1$, we define the Wronskian W as

$$W(f_1, \ldots, f_n)[t] = \det \begin{pmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \vdots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{pmatrix} [t].$$

General theory

Theorem 11.13

(Abel's theorem). *Let $y_1, \ldots, y_n$ be solutions to the homogeneous equation*

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = 0,$$

for $t \in I$. Then,

$$W(y_1, \ldots, y_n)[t] = ce^{-\int p_{n-1}(t)dt}$$

for a constant c not dependent on $t \in I$.

High-order linear homogeneous ODE

The homogeneous linear ODE is of the form

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y =,$$

Theorem 11.14

If $p_0, \ldots, p_{n-1}$ are continuous functions in I, and $y_1, \ldots, y_n$ are solutions to the above homogeneous equation, then every solution ϕ to the homogeneous equation can be expressed as a linear combination of $y_1, \ldots, y_n$ if and only if $W(y_1, \ldots, y_n)[t_0] \neq 0$ for some $t_0 \in I$. In this case, we call $(y_1, \ldots, y_n)$ a **fundamental set of solutions (FSS)** *to the homogeneous equation.*

Homogeneous equation with constant coefficients

For

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0.$$

A trial function $\phi = e^{rt}$ for $r \in \mathbb{R}$. The **characteristic equation** is

$$a_n r^n + \cdots + a_1 r + a_0 = 0.$$

High-order non-homogeneous equation with constant coefficients

$$a_n y^n + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(t). \tag{22}$$

(1) $g(t) = e^{\alpha t} P_m(t)$,

(2) $g(t) = e^{\alpha t} P_m(t) \cos(\beta t)$, or $g(t) = e^{\alpha t} P_m(t) \sin(\beta t)$.

Method of undetermined coefficients

The possible particular solutions can be used are

(1) $Y(t) = t^s e^{\alpha t} Q_m(t)$,where

$$Q_m(t) = A_m t^m + \cdots + A_1 t + A_0$$

for undetermined coefficients $A_m, \ldots, A_0$, and

$$s = \begin{cases} 0, & \text{if } \alpha \text{ is not a root of the characteristic equation (*).} \\ m, & \text{if } \alpha \text{ is a root of the characteristic equation (*) with multiplicity } m \end{cases}$$

(2) $Y(t) = t^s e^{\alpha t}[Q_m(t)\cos(\beta t) + R_m(t)\sin(\beta t)]$, where $Q_m = A_m t^m + \cdots + A_1 t + A_0$, $R_m = B_m t^m + \cdots + B_1 t + B_0$ are polynomials of degree m with undetermined coefficients $A_m, \ldots, A_0, B_m, \ldots, B_0$, and

$$s = \begin{cases} 0, & \text{if } \alpha + i\beta \text{ is not a root of the characteristic equation (*).} \\ m, & \text{if } \alpha + i\beta \text{ is a root of the characteristic equation(*) with multiplicity } m. \end{cases}$$

Variation of parameters-high-order non-homogeneous ODE

Similar to second order equations, there is also a method to treat rather general high order equations

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = g(t), \quad t \in I.$$

Suppose we have a fundamental set of solutions to $y_1, \ldots, y_n$ to the homogeneous equation. Then, the complementary solution is

$$y_c(t) = c_1 y_1(t) + \cdots + c_n y_n(t).$$

Now, we consider a trial solution for the non-homogeneous equation of the form

$$Y(t) = u_1(t)y_1(t) + \cdots + u_n(t)y_n(t)$$

for unknown functions $u_1, \ldots, u_n$. Imposing

$$u_1'(t)y_1^{(m)}(t) + \cdots + u_n'(t)y_n^{(m)}(t) = 0 \quad \forall 1 \leq m \leq n-2,$$

one has

Variation of parameters

$$\begin{pmatrix} y_1 & y_2 & \cdots & y_{n-1} & y_n \\ y_1' & y_2' & \cdots & y_{n-1}' & y_n' \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_{n-1}^{(n-2)} & y_n^{(n-2)} \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_{n-1}^{(n-1)} & y_n^{(n-1)} \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ \vdots \\ u_{n-1}' \\ u_n' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ g(t) \end{pmatrix}.$$

Variation of parameters

To invert $M(t)$, we use Cramers rule, by setting

$$M_i(t) = \begin{pmatrix} y_1 & \dots & 0 & \dots & y_n \\ y_1' & \dots & 0 & \dots & y_n' \\ \vdots & & \vdots & & \vdots \\ y_1^{(n-2)} & \dots & 0 & \dots & y_n^{(n-2)} \\ y_1^{(n-1)} & \dots & 1 & \dots & y_n^{(n-1)} \end{pmatrix},$$

i.e., replace the ith column of $M(t)$ with the vector $(0, \dots, 0, 1)^T$. Then Cramer's rule gives

$$u_i'(t) = \frac{g(t) \det M_i(t)}{\det M(t)},$$

and by integrating we get an expression for $u_i(t)$. The particular solution to the non-homogeneous equation is therefore

$$Y(t) = y_1(t) \int \frac{g(t) \det M_1(t)}{\det M(t)} dt + \cdots + y_n(t) \int \frac{g(t) \det M_n(t)}{\det M(t)} dt.$$

First order ODE systems

Definition 11.15

If each $F_i, 1 \leq i \leq n$, is linear with respect to $y_1, \ldots, y_n$, then we call the system of ODEs a linear system. Otherwise it is a nonlinear system.
We now study linear systems is greater detail. The general linear system of first order ODEs is

$$\begin{aligned}
y_1'(t) &= p_{11}(t)y_1(t) + p_{12}(t)y_2(t) + \cdots + p_{1n}(t)y_n(t) + g_1(t), \\
y_2'(t) &= p_{21}(t)y_1(t) + p_{22}(t)y_2(t) + \cdots + p_{2n}(t)y_n(t) + g_2(t), \\
&\vdots \\
y_n'(t) &= p_{n1}(t)y_1(t) + p_{n2}(t)y_2(t) + \cdots + p_{nn}(t)y_n(t) + g_n(t),
\end{aligned}$$

where $p_{11}(t), \ldots, p_{nn}(t), g_1(t), \ldots, g_n(t)$ are given real functions.

First order linear ODE systems

Definition 11.16

It is more convenient to introduce the matrix form. Denoting vectors

$$\mathbf{y}(t) = (y_1(t), \ldots, y_n(t))^T, \quad \mathbf{g}(t) = (g_1(t), \ldots, g_n(t))^T,$$

and the matrix

$$\mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

the general system can be written as

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t).$$

First order linear ODE systems

The above linear first-order ODE system subject to the following initial conditions

$$y_1(t_0) = x_1, \quad \ldots, \quad y_n(t_0) = x_n,$$

is the initial value problem (IVP).

Theorem 11.17

(Existence and uniqueness for linear systems). *Let $I = (\alpha, \beta) \subset \mathbb{R}$ be an open interval such that functions $p_{11}(t), \ldots, p_{nn}(t), g_1(t), \ldots, g_n(t)$ are continuous in I. For $t_0 \in I$ and $x_1, \ldots, x_n \in \mathbb{R}$, there is exactly one solution $\mathbf{y} = (y_1(t), \ldots, y_n(t))^T$ to the IVP.*

Wronskian

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \tag{23}$$

Definition 11.18

(Wronskian). Let $\mathbf{y}_1(t), \ldots, \mathbf{y}_n(t)$ be n solutions to the homogeneous system (24). We define the matrix

$$\mathbf{X}(t) := \begin{pmatrix} y_{11}(t) & y_{12}(t) & \ldots & y_{1n}(t) \\ y_{21}(t) & y_{22}(t) & \ldots & y_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1}(t) & y_{n2}(t) & \ldots & y_{nn}(t) \end{pmatrix} = \begin{pmatrix} | & | & \ldots & | \\ \mathbf{y}_1(t) & \mathbf{y}_2(t) & \ldots & \mathbf{y}_n(t) \\ | & | & \ldots & | \end{pmatrix},$$

where the i-th column of $\mathbf{X}$ is the vector $\mathbf{y}_i(t)$. Then we set the Wronskian $W(\mathbf{y}_1, \ldots, \mathbf{y}_n)[t]$ to be

$$W(\mathbf{y}_1, \ldots, \mathbf{y}_n)[t] := \det \mathbf{X}(t).$$

Fundamental solution set

Theorem 11.19

Let $\mathbf{y}_1(t), \ldots, \mathbf{y}_n(t)$ *be* n *solutions to the homogenous system (24) defined on an open interval* I*. Then,* $\mathbf{y}_1(t), \ldots, \mathbf{y}_n(t)$ *are linearly independent at each point in the interval* I **if and only if** *the Wronskian* $W(\mathbf{y}_1, \ldots, \mathbf{y}_n)[t]$ *is non-zero for* $t \in I$*. In this case, we say that* $\{\mathbf{y}_1(t), \ldots, \mathbf{y}_n(t)\}$ *forms a fundamental set of solutions (FSS), and any solution* $\phi(t)$ *to the homogeneous system (24) can be expressed as a linear combination:*

$$\phi(t) = c_1\mathbf{y}_1(t) + \cdots + c_n\mathbf{y}_n(t),$$

for constants $c_1, \ldots, c_n \in \mathbb{R}$ *in* **exactly one way**. *That is, the constants* $c_1, \ldots, c_n$ *are* **uniquely** *determined.*

Liouville's formula

Since we have an analogue of the Wronskian for system of equations, we should expect an analogue of Abel's theorem as well. For systems of equations, this is called **Liouville's formula**.

Theorem 11.20

(Liouville's formula). *Let $\mathbf{y}_1, \ldots, \mathbf{y}_n$ be n solutions to the homogeneous equation (24) in the open interval I. Then, the Wronskian is given by*

$$W(\mathbf{y}_1, \ldots, \mathbf{y}_n)[t] = c \exp\left(\int tr(\mathbf{P}(t)) dt \right),$$

where the trace of a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is defined as

$$tr(\mathbf{A}) := \sum_{i=1}^{n} a_{ii} \qquad \text{(sum of the diagonal entries)},$$

and c is a constant not depending on $t \in I$. Consequently, the Wronskian is either always zero for $t \in I$ or never zero for $t \in I$.

Complex-valued solution

Just as for second order equations, you will see that a linear ODE system with real-valued coefficients may give rise to complex-valued solutions. But again we also have the following theorem.

Theorem 11.21

If $\mathbf{y}(t) = \mathbf{u}(t) + i\mathbf{v}(t)$ *is a complex-valued solution to the homogeneous system (24), where the entries of* $\mathbf{P}(t)$ *are* **real-valued** *functions, and the vectors* $\mathbf{u}(t)$ *and* $\mathbf{v}(t)$ *are also real-valued, then* $\mathbf{u}(t)$ *and* $\mathbf{v}(t)$ *are both solutions to the homogeneous system (24).*

Two methods for finding the FSS

For first-order linear ODE system

$$\mathbf{y}' = \mathbf{A}\mathbf{y}(t)$$

There are two methods to get its FSS.

- Eigenvalue and generalized eigenvectors.
- e^{At}

Diagonalizable matrix

Recall:

Theorem

Let A be a square matrix with size n, then A is diagonalizable if and only if the algebraic multiplicity and geometric multiplicity are the same for each eigenvalue.

If $q_i = m_i$ for all $i = 1, \ldots, k$, then $\mathbf{A}$ is diagonalizable. There are n linearly independent eigenvectors $\xi_1, \cdots, \xi_n$ corresponding to eigenvalues $r_1, \cdots, r_n$, where $r_1, \cdots, r_n \in \{\lambda_1, \ldots, \lambda_k\}$, and
$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{\Lambda} = diag(r_1, \cdots, r_n), \mathbf{P} = [\xi_1, \cdots, \xi_n]$.
We define the new vector $\mathbf{x} := \mathbf{P}^{-1}\mathbf{y}$. Then,

$$\mathbf{x}'(t) = \mathbf{P}^{-1}\mathbf{y}'(t) = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{x}(t) \Rightarrow \boxed{\mathbf{x}'(t) = \mathbf{\Lambda}\mathbf{x}(t)}.$$

The solution for $\mathbf{x}$ can be solved easily, which is given by:

$$\mathbf{x}(t) = [c_1 e^{r_1 t}, \cdots, c_n e^{r_n t}]^T.$$

and $\mathbf{y}(t) = \mathbf{P}\mathbf{x}(t) = c_1 e^{r_1 t}\xi_1 + \cdots + c_n e^{r_n t}\xi_n$.
Indeed, $e^{r_1 t}\xi_1, \cdots, e^{r_n t}\xi_n$ is a fundamental set of solutions.

Eigenvalue and generalized eigenvectors for non-diagonalizable matrix

If there is a repeated eigenvalue λ with geometric multiplicity **strictly less** than its algebraic multiplicity, $\mathbf{A}$ is not diagonalizable. In this case, it is more complicated.

In the following, we carry out a systematic way to find m_i linearly independent solutions corresponding to the eigenvalue λ_i (for $i = 1, \cdots, k$), where $m_1 + \cdots + m_k = n$.

The general case: $n \times n$ matrix

For all distinct eigenvalues $(\lambda_1, \cdots, \lambda_k)$, we will need to carry out the following process.

For $i = 1, \cdots, k$, do the following.

Let $\lambda = \lambda_i$ be the eigenvalue of $\mathbf{A}$, $m = m_i$ is algebraic multiplicity of λ, and $q = q_i$ is geometric multiplicity of λ. Then we want to find m linearly independent solutions corresponding to the eigenvalue λ.

Case 1: If geometric multiplicty q=algebraic multiplicity m for the eigenvalue λ, then suppose $\mathbf{r}_1, \cdots, \mathbf{r}_m$ are m linearly independent eigenvectors w.r.t λ $(\mathbf{A}\mathbf{r}_j = \lambda \mathbf{r}_j, j = 1, \cdots, m)$. $(\frac{d}{dt}\left(\mathbf{r}_j e^{\lambda t}\right) = \lambda \mathbf{r}_j e^{\lambda t})$
Thus, we already have m linearly independent solutions $\mathbf{r}_1 e^{\lambda t}, \cdots, \mathbf{r}_m e^{\lambda t}$.

Case: geometric multiplicty q <algebraic multiplicity m

Case 2: If geometric multiplicty $q <$ algebraic multiplicity m for the eigenvalue λ. Then we will need to construct m linearly independent solutions. We look for the solutions of the following form:

$$\mathbf{y}(t) = \left(\mathbf{r}_0 + \mathbf{r}_1 t + \mathbf{r}_2 \frac{t^2}{2} + \cdots + \mathbf{r}_{m-1} \frac{t^{m-1}}{(m-1)!}\right) e^{\lambda t}$$

Then

$$\begin{aligned}
&\mathbf{y}'(t) - \mathbf{A}\mathbf{y}(t) \\
&= \left(\mathbf{r}_1 + \mathbf{r}_2 t + \cdots + \mathbf{r}_{m-1} \frac{t^{m-2}}{(m-2)!}\right) e^{\lambda t} \\
&+ \left(\lambda \mathbf{r}_0 + \lambda \mathbf{r}_1 t + \cdots + \lambda \mathbf{r}_{m-1} \frac{t^{m-1}}{(m-1)!}\right) e^{\lambda t} \\
&- \left(A\mathbf{r}_0 + A\mathbf{r}_1 t + \cdots + A\mathbf{r}_{m-1} \frac{t^{m-1}}{(m-1)!}\right) e^{\lambda t} \\
&= (\mathbf{r}_1 - (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_0) e^{\lambda t} + (\mathbf{r}_2 - (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_1) t e^{\lambda t} + (\mathbf{r}_3 - (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_2) \frac{t^2}{2} e^{\lambda t} + \cdots \\
&+ (\mathbf{r}_{m-1} - (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_{m-2}) \frac{t^{m-2}}{(m-2)!} e^{\lambda t} - (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_{m-1} \frac{t^{m-1}}{(m-1)!} e^{\lambda t}
\end{aligned}$$

Case: geometric multiplicty q <algebraic multiplicity m

In order to get $\mathbf{y}'(t) = A\mathbf{y}(t)$, we need

$$\begin{aligned}
\mathbf{r}_1 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_0 \\
\mathbf{r}_2 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_1 \\
\mathbf{r}_3 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_2 \\
&\vdots \\
\mathbf{r}_{m-1} &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_{m-2} \\
(\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_{m-1} &= 0.
\end{aligned}$$

Substituting the first $m-1$ equations into the last equation gives

$$\begin{aligned}
(A-\lambda I)^m\mathbf{r}_0 &= 0 \\
\mathbf{r}_1 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_0 \\
\mathbf{r}_2 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_1 \\
\mathbf{r}_3 &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_2 \\
&\vdots \\
\mathbf{r}_{m-1} &= (\mathbf{A} - \lambda\mathbf{I})\mathbf{r}_{m-2}.
\end{aligned}$$

Case: geometric multiplicty q <algebraic multiplicity m

Fact

If λ is an eigenvalue of $\mathbf{A}$ with algebraic multiplicity m, then

$$(\mathbf{A} - \lambda \mathbf{I})^m \mathbf{r} = \mathbf{0}$$

will have m linearly independent solutions $\mathbf{r}_0^{(1)}, \cdots, \mathbf{r}_0^{(m)}$.

Remark: $Null((\mathbf{A} - \lambda \mathbf{I})^m)$ is called the generalized eigenspace for eigenvalue λ with algebraic multiplicity m, $\dim(Null((\mathbf{A} - \lambda \mathbf{I})^m)) = m$. We skip the proof. (It can be proved by using the Jordan matrix form of $\boldsymbol{A}$)
And this fact have been proved in many Advanced linear algebra textbooks.

Case: geometric multiplicty q <algebraic multiplicity m

Start from $\mathbf{r}_0^{(1)}$, we have

$$\begin{aligned}
\mathbf{r}_1^{(1)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_0^{(1)} \\
\mathbf{r}_2^{(1)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_1^{(1)} \\
\mathbf{r}_3^{(1)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_2^{(1)} \\
&\vdots \\
\mathbf{r}_{m-1}^{(1)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_{m-2}^{(1)}.
\end{aligned}$$

We can construct the first solution

$$\mathbf{y}^{(1)}(t) = \left(\mathbf{r}_0^{(1)} + \mathbf{r}_1^{(1)} t + \mathbf{r}_2^{(1)} \frac{t^2}{2} + \cdots + \mathbf{r}_{m-1}^{(1)} \frac{t^{m-1}}{(m-1)!} \right) e^{\lambda t}$$

Case: geometric multiplicty q <algebraic multiplicity m

Start from $\mathbf{r}_0^{(2)}$, we have

$$\begin{aligned}
\mathbf{r}_1^{(2)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_0^{(2)} \\
\mathbf{r}_2^{(2)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_1^{(2)} \\
\mathbf{r}_3^{(2)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_2^{(2)} \\
&\vdots \\
\mathbf{r}_{m-1}^{(2)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_{m-2}^{(2)}.
\end{aligned}$$

We can construct the second solution

$$\mathbf{y}^{(2)}(t) = \left(\mathbf{r}_0^{(2)} + \mathbf{r}_1^{(2)} t + \mathbf{r}_2^{(2)} \frac{t^2}{2} + \cdots + \mathbf{r}_{m-1}^{(2)} \frac{t^{m-1}}{(m-1)!} \right) e^{\lambda t}$$

Case: geometric multiplicty q <algebraic multiplicity m

The process continue until we start from $\mathbf{r}_0^{(m)}$, then we have

$$\begin{aligned}
\mathbf{r}_1^{(m)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_0^{(m)} \\
\mathbf{r}_2^{(m)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_1^{(m)} \\
\mathbf{r}_3^{(m)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_2^{(m)} \\
&\vdots \\
\mathbf{r}_{m-1}^{(m)} &= (\mathbf{A} - \lambda \mathbf{I})\mathbf{r}_{m-2}^{(m)}.
\end{aligned}$$

We can construct the mth solution

$$\mathbf{y}^{(m)}(t) = \left(\mathbf{r}_0^{(m)} + \mathbf{r}_1^{(m)} t + \mathbf{r}_2^{(m)} \frac{t^2}{2} + \cdots + \mathbf{r}_{m-1}^{(m)} \frac{t^{m-1}}{(m-1)!} \right) e^{\lambda t}$$

Case: geometric multiplicty q <algebraic multiplicity m

Fact

$$\mathbf{y}^{(1)}(t), \cdots, \mathbf{y}^{(m)}(t)$$

are m linearly independent solutions corresponding to the eigenvalue λ for any $t \in \mathbb{R}$.

Fundamental matrices and matrix exponential

Recall:

$$\mathbf{x}'(t) = \mathbf{P}(t)\mathbf{x}(t) \tag{24}$$

Definition 11.22

Suppose that $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ form a fundamental set of solutions for the homogeneous linear system (24). Then the matrix

$$\mathbf{\Psi}(t) = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{x}^{(1)}(t) & \mathbf{x}^{(2)}(t) & \cdots & \mathbf{x}^{(n)}(t) \\ | & | & \cdots & | \end{pmatrix} \tag{25}$$

whose columns are the vectors $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ is called a **fundamental matrix** of the system (24).

Fundamental matrix

Propery:
Let $\mathbf{y}_1(t), \ldots, \mathbf{y}_n(t)$ be a fundamental set of solutions to the homogeneous system $\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t)$. The matrix $\mathbf{F}$ defined as

$$\mathbf{F}(t) = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{y}_1(t) & \mathbf{y}_2(t) & \cdots & \mathbf{y}_n(t) \\ | & | & \cdots & | \end{pmatrix}$$

satisfies the property

$$\frac{d\mathbf{F}(t)}{dt} = \mathbf{P}(t)\mathbf{F}(t)$$

Matrix exponential.

$$e^{\mathbf{At}} \triangleq \mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A^n t^n}}{n!}. \tag{26}$$

Theorem 11.23

$$\frac{d(e^{\mathbf{A}t})}{dt} = (e^{\mathbf{A}t})' = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}. \tag{27}$$

Fact

$e^{\mathbf{A}t}$ is the fundamental matrix for the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

with $e^{\mathbf{A}t}|_{t=0} = I$.

Matrix exponential.

Property

1. $\exp(\mathbf{O}_{n\times n}) = \boldsymbol{I}$.
2. If $\Lambda = diag(\lambda_1, \cdots, \lambda_n)$, then $e^{\Lambda} = diag(e^{\lambda_1}, \cdots, e^{\lambda_n})$.

Theorem 11.24

1. *If* $\boldsymbol{AB} = \boldsymbol{BA}$, *then* $e^{\boldsymbol{A}+\boldsymbol{B}} = e^{\boldsymbol{A}}e^{\boldsymbol{B}}$.
2. $(e^{\boldsymbol{A}})^{-1} = e^{-\boldsymbol{A}}$
3. *For non-singular matrix* $\boldsymbol{P}$, *we have*

$$\exp(\boldsymbol{PAP}^{-1}) = \boldsymbol{P}e^{\boldsymbol{A}}\boldsymbol{P}^{-1}.$$

Theorem 11.25

If $\mathbf{A}$ *is diagonalizable, then there exists an invertible matrix* $\mathbf{P}$ *such that* $\boldsymbol{A} = \boldsymbol{P}diag(\lambda_1, \cdots, \lambda_n)\boldsymbol{P}^{-1}$, *then* $e^{\mathbf{A}} = \boldsymbol{P}diag(e^{\lambda_1}, \cdots, e^{\lambda_n})\boldsymbol{P}^{-1}$.

Semisimple-Nilpotent decomposition (S-N decomposition)

Definition 11.26

A square matrix is called Semisimple if it is diagonalizable.

And a square matrix A is called Nilpotent if there is some positive integer k s.t. $A^k = O$.

S-N decomposition (Semisimple-Nilpotent decomposition)

Theorem 11.27

(Semisimple-Nilpotent decomposition)
Let $\boldsymbol{A}$ be an $n \times n$ matrix. Then, there exist two $n \times n$ matrices $\boldsymbol{S}$ and $\boldsymbol{N}$ such that
(a) $\boldsymbol{S}$ is diagonalizable (semisimple),
(b) $\boldsymbol{N}$ is Nilpotent,
(c) $\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N}$,
(d) $\boldsymbol{SN} = \boldsymbol{NS}$.
The two matrices $\boldsymbol{S}$ and $\boldsymbol{N}$ are uniquely determined by these four conditions.

We skip the proof for this theorem, you could check the book "*Basic Theory of Ordinary Differential Equations, Po-Fang Hsieh, Yasutaka Sibuya, Springer, 1999*" .

S-N decomposition

Although $S - N$ decomposition exists for any $n \times n$ matrix, but the construction of S and N for a general $n \times n$ matrix is related to Jordan form of the matrix (I think most students don't known). Thus, for simplicity, in this course, we only focus on the $S - N$ decomposition for 2×2 and 3×3 matrices with one single eigenvalue:

Theorem 11.28 (S-N decomposition)

Let $\boldsymbol{A}$ be a 2×2 or 3×3 matrix which has only one distinct eigenvalue r. Then $\boldsymbol{A}$ could be decomposed as $\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N}$ such that

1. $\boldsymbol{S} = r\boldsymbol{I}$.
2. $\boldsymbol{N} = \boldsymbol{A} - \boldsymbol{S}$.
3. $\boldsymbol{N}^2 = \boldsymbol{O}$ *or* $\boldsymbol{N}^3 = \boldsymbol{O}$.
4. $\boldsymbol{SN} = \boldsymbol{NS}$

We also skip the proof for this theorem, you could check the book "*Basic Theory of Ordinary Differential Equations, Po-Fang Hsieh, Yasutaka Sibuya, Springer, 1999*" .

S-N decomposition for 3×3 matrix

Example 11.29

The general solution is given by

$$\boldsymbol{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -3 & 2 & 4 \end{pmatrix} \boldsymbol{x}$$

The eigenvalues of $\boldsymbol{A}$ must satisfy:

$$\det(\boldsymbol{A} - r\boldsymbol{I}) = \begin{vmatrix} 1-r & 1 & 1 \\ 2 & 1-r & -1 \\ -3 & 2 & 4-r \end{vmatrix} = -(r-2)^3 \implies r = 2.$$

We perform the S-N decompositon for $\boldsymbol{A}$:

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} + \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{pmatrix}$$

S-N decomposition for 3×3 matrix

Example continue

And we observe

$$e^{\boldsymbol{S}t} = e^{2t} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$\boldsymbol{N}^2 = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix}, \boldsymbol{N}^3 = \boldsymbol{O}$$

It follows that

$$e^{\boldsymbol{N}t} = \boldsymbol{I} + \boldsymbol{N}t + \frac{1}{2}\boldsymbol{N}^2 t^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -t & t & t \\ 2t & -t & -t \\ -3t & 2t & 2t \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ -\frac{t^2}{2} & \frac{t^2}{2} & \frac{t^2}{2} \\ \frac{t^2}{2} & -\frac{t^2}{2} & -\frac{t^2}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1-t & t & t \\ 2t - \frac{t^2}{2} & 1 - t + \frac{t^2}{2} & -t + \frac{t^2}{2} \\ -3t + \frac{t^2}{2} & 2t - \frac{t^2}{2} & 1 + 2t - \frac{t^2}{2} \end{pmatrix}.$$

S-N decomposition for 3×3 matrix

Example continue

The fundamental matrix is given by:

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t}e^{\boldsymbol{N}t} = e^{2t}\begin{pmatrix} 1-t & t & t \\ 2t-\frac{t^2}{2} & 1-t+\frac{t^2}{2} & -t+\frac{t^2}{2} \\ -3t+\frac{t^2}{2} & 2t-\frac{t^2}{2} & 1+2t-\frac{t^2}{2} \end{pmatrix}$$

$$e^{\boldsymbol{A}t}(t=0) = I.$$

Thus the general solution is given by:

$$\boldsymbol{x} = c_1\begin{pmatrix} 1-t \\ 2t-\frac{t^2}{2} \\ -3t+\frac{t^2}{2} \end{pmatrix}e^{2t} + c_2\begin{pmatrix} t \\ 1-t+\frac{t^2}{2} \\ 2t-\frac{t^2}{2} \end{pmatrix}e^{2t} + c_3\begin{pmatrix} t \\ -t+\frac{t^2}{2} \\ 1+2t-\frac{t^2}{2} \end{pmatrix}e^{2t}.$$

Method of undetermined coefficients-non-homogeneous system

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t) + \mathbf{g}(t).$$

We now list the trial solutions for specific examples of $\mathbf{g}$:

$\mathbf{g}(t)$	Particular solution form $\mathbf{Y}(t)$	value of s
$\mathbf{P}_m(t)e^{\alpha t}$	$\mathbf{Q}_{m+s}(t)e^{\alpha t}$	alg. mult. of α
$\mathbf{P}_m(t)e^{\alpha t}\cos(\beta t)$	$\mathbf{Q}_{m+s}(t)e^{\alpha t}\cos(\beta t) + \mathbf{R}_{m+s}(t)e^{\alpha t}\sin(\beta t)$	alg. mult. of $\alpha + i\beta$
$\mathbf{P}_m(t)e^{\alpha t}\sin(\beta t)$	$\mathbf{Q}_{m+s}(t)e^{\alpha t}\cos(\beta t) + \mathbf{R}_{m+s}(t)e^{\alpha t}\sin(\beta t)$	alg. mult. of $\alpha + i\beta$

alg. mult. of α means the multiplicity of α as the eigenvalue of $\mathbf{A}$.

alg. mult. of $\alpha + i\beta$ means the multiplicity of $\alpha + i\beta$ as the eigenvalue of $\mathbf{A}$.

Method of undetermined coefficients-non-homogeneous system

Remark 1

In constrast to n-th order linear equations, where the form of the trial solution is $t^s \mathbf{Q}_m(t)$, so that the lowest order term is t^s, for linear systems we have to use a trial solution of the form $\mathbf{Q}_{m+s}(t)$, which is a polynomial of degree $m+s$ which includes all lower order terms $t^{s-1}, t^{s-2}, \ldots, t^1, t^0$.

Variation of parameters-non-homogeneous system

Returning to the non-homogeneous system

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t).$$

Assume we have a fundamental matrix $\mathbf{F}(t)$ to the homogeneous system with complementary solution

$$\mathbf{y}_c(t) = \mathbf{F}(t)\mathbf{c},$$

where $\mathbf{c}$ is a constant vector. The method of variation of parameters is to consider a trial solution

$$\mathbf{z}(t) = \mathbf{F}(t)\mathbf{u}(t),$$

where $\mathbf{u}(t)$ is a vector of functions. Then, if $\mathbf{z}$ is a solution to the non-homogeneous system, we find that

$$\mathbf{P}(t)\mathbf{F}(t)\mathbf{u}(t) + \mathbf{g}(t) = \mathbf{z}'(t) = \mathbf{F}(t)\mathbf{u}'(t) + \mathbf{F}'(t)\mathbf{u}(t).$$

Since $\mathbf{F}(t)$ is a fundamental matrix, i.e., $\mathbf{F}'(t) = \mathbf{P}(t)\mathbf{F}(t)$, we see that

$$\mathbf{F}(t)\mathbf{u}'(t) = \mathbf{g}(t) \Rightarrow \mathbf{u}'(t) = \mathbf{F}^{-1}(t)\mathbf{g}(t).$$

Variation of parameters-non-homogeneous system

Integrating this gives

$$\mathbf{u}(t) = \int \mathbf{F}^{-1}(t)\mathbf{g}(t)dt.$$

The particular solution is

$$\mathbf{z}(t) = \mathbf{F}(t) \int \mathbf{F}^{-1}(t)\mathbf{g}(t)dt.$$

Therefore the general solution to the non-homogeneous system is

$$\mathbf{y}(t) = \mathbf{y}_c(t) + \mathbf{z}(t) = \mathbf{F}(t)\mathbf{c} + \mathbf{F}(t) \left[\int \mathbf{F}^{-1}(t)\mathbf{g}(t)dt \right].$$

Variation of parameters-non-homogeneous system

If we are also given initial conditions $\mathbf{y}(t_0) = \mathbf{v}$, then in the integral we write

$$\int_{t_0}^{t} \mathbf{F}^{-1}(s)\mathbf{g}(s)ds$$

so that

$$\mathbf{v} = \mathbf{y}(t_0) = \mathbf{F}(t_0)\mathbf{c} \Rightarrow \mathbf{c} = \mathbf{F}^{-1}(t_0)\mathbf{v}.$$

Hence, the unique solution to the IVP in the interval I is

$$\mathbf{y}(t) = \mathbf{F}(t)\mathbf{F}^{-1}(t_0)\mathbf{v} + \mathbf{F}(t)\left[\int_{t_0}^{t} \mathbf{F}^{-1}(s)\mathbf{g}(s)ds\right].$$

Definition for Laplace Transform

Definition 11.30

Suppose that $f(t)$ is a real function defined on $[0, +\infty)$ and

1. f is piecewise continuous on the interval $0 \leq t \leq A$ for any positive A.
2. $|f(t)| \leq Ke^{at}$ when $t \geq M$. In this inequality, K, a, and M are real constants ($K > 0, M > 0$). ($f(t)$ is in the "exponential" order.)

Then the **Laplace transform** $\mathcal{L}\{f(t)\}(s) = F(s)$ defined as

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt, \tag{28}$$

exists for $s > a$.
Moreover, in this case, we can define **inverse Laplace transform** for $F(s) = \mathcal{L}\{f(t)\}(s > a)$ as

$$\mathcal{L}^{-1}\{F(s)\} = f(t). \tag{29}$$

Definition for Laplace Transform

Theorem 11.31

Suppose that f is continuous and f' is piecewise continuous on any interval $0 \le t \le A$. Suppose further that there exist constants K, a, and M such that $|f(t)| \le Ke^{at}$ for $t \ge M$. Then $\mathcal{L}\{f'(t)\}$ exists for $s > a$, and moreover,

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0). \tag{30}$$

Theorem 11.32

Suppose that f is continuous and f' is piecewise continuous on any interval $0 \le t \le A$. Suppose further that there exist constants K, a, and M such that $|f(t)| \le Ke^{at}$ for $t \ge M$. Then $\mathcal{L}\{f'(t)\}$ exists for $s > a$, and moreover,

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0). \tag{31}$$

Definition for Laplace Transform

Corollary

Suppose that $f, f', \cdots, f^{(n-1)}$ *are continuous and* $f^{(n)}$ *is piecewise continuous on any interval* $0 \le t \le A$. *Suppose further that there exist constants* K, a *and* M *such that* $|f^{(i)}(t)| \le Ke^{at}$, $i = 0, 1, \cdots, n-1$ *for* $t \ge M$. *Then* $\mathcal{L}\{f^{(n)}(t)\}$ *exists for* $s > a$, *and*

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - \cdots - f^{(n-1)}(0). \qquad \square \tag{32}$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Theorem 11.33

If $F(s) = \mathcal{L}\{f(t)\}$ *exists for* $s > a$, *and if* c *is a constant, then*

$$\mathcal{L}\{e^{ct} f(t)\} = F(s - c), \quad s > a + c. \tag{33}$$

Conversely, if $f(t) = \mathcal{L}^{-1}\{F(s)\}$, *then*

$$e^{ct} f(t) = \mathcal{L}^{-1}\{F(s - c)\}. \tag{34}$$

Definition 11.34

The **unit step function** *or* **Heaviside function** *is defined by*

$$u_c(t) = \begin{cases} 0, & t < c, \\ 1, & t \geq c, \end{cases} \qquad c \geq 0. \qquad \square \tag{35}$$

Theorem 11.35

If $F(s) = \mathcal{L}\{f(t)\}$ *exists for* $s > a$*, and if* c *is a positive constant, then*

$$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}\mathcal{L}\{f(t)\} = e^{-cs}F(s), \quad s > a. \tag{36}$$

Conversely, if $f(t) = \mathcal{L}^{-1}\{F(s)\}$*, then*

$$u_c(t)f(t-c) = \mathcal{L}^{-1}\{e^{-cs}F(s)\}. \tag{37}$$

Unit impulses and the Dirac delta functions

Suppose $g(t)$ is given by

$$g(t) = d_\tau(t) = \begin{cases} \dfrac{1}{2\tau}, & -\tau < t < \tau, \\ 0, & t \le -\tau, \text{ or } t \ge \tau, \end{cases} \tag{38}$$

where $\tau > 0$ is a small constant. In this case, it follows immediately that, $I(\tau) = 1$ independent of the value of τ, as long as $\tau \neq 0$.

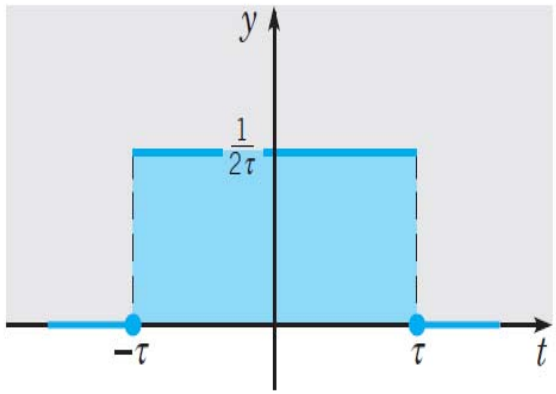
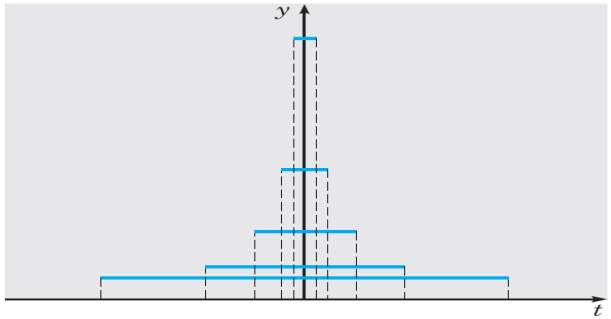


Fig 6. Graph of $y = d_\tau(t)$.

Unit impulses and the Dirac delta functions

In the limit as $\tau \to 0$, we get an idealized unit impulse function δ, which imposes an impulse of magnitude one at $t = 0$ but is zero for all t other than zero. This function, which is not an ordinary function studied in elementary calculus, is called Dirac **delta function** .



Graph of $y = d_\tau(t)$ as $\tau \to 0^+$.

Unit impulses and the Dirac delta functions

The delta function has the following properties:

$$\delta(t) = 0, \ t \neq 0, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t)dt = 1. \tag{39}$$

A unit impulse at an arbitrary point $t = t_0$ is given by $\delta(t - t_0)$, which satisfies

$$\delta(t - t_0) = 0, \ t \neq t_0, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - t_0)dt = 1. \tag{40}$$

Important property: Dirac delta function $\delta(t - t_0)$ satisfies the following property: for any integrable function $f(t)$ over $(-\infty, +\infty)$,

$$\lim_{\tau \to 0^+} \int_{-\infty}^{\infty} d_\tau(t - t_0)f(t)dt = \int_{-\infty}^{\infty} \delta(t - t_0)f(t)dt. \tag{41}$$

Unit impulses and the Dirac delta functions

Theorem 11.36

Suppose that $f(t)$ is integrable on $(-\infty, +\infty)$ and continuous near t_0. Then

$$\int_{-\infty}^{\infty} \delta(t - t_0) f(t) dt = f(t_0). \qquad \square \tag{42}$$

For $t_0 > 0$, the Laplace transform of $\delta(t - t_0)$ is defined by

$$\begin{aligned}
\mathcal{L}\{\delta(t - t_0)\} &= \lim_{\tau \to 0+} \mathcal{L}\{d_\tau(t - t_0)\} \\
&= \lim_{\tau \to 0+} \int_0^{\infty} e^{-st} d_\tau(t - t_0) dt \\
&= \lim_{\tau \to 0+} \int_{t_0 - \tau}^{t_0 + \tau} e^{-st} d_\tau(t - t_0) dt \\
&= \lim_{\tau \to 0+} \frac{1}{2\tau} \int_{t_0 - \tau}^{t_0 + \tau} e^{-st} dt \\
&= \lim_{\tau \to 0+} \frac{1}{2s\tau} e^{-st_0}(e^{s\tau} - e^{-s\tau}) = e^{-st_0}
\end{aligned}$$

Laplace Transform Table

Table for Elementary Laplace Transforms

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s},\ s > 0$
2.	e^{at}	$\frac{1}{s-a},\ s > a$
3.	$t^n,\ n > 0$ integer	$\frac{n!}{s^{n+1}},\ s > 0$
4.	$t^p,\ p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}},\ s > 0$
5.	$\sin at$	$\frac{a}{s^2+a^2},\ s > 0$
6.	$\cos at$	$\frac{s}{s^2+a^2},\ s > 0$
7.	$\sinh at$	$\frac{a}{s^2-a^2},\ s > \lvert a\rvert$
8.	$\cosh at$	$\frac{s}{s^2-a^2},\ s > \lvert a\rvert$

Laplace Transform Table

Table for Elementary Laplace Transforms

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
9.	$e^{at}\sin bt$	$\dfrac{b}{(s-a)^2+b^2},\ s>a$
10.	$e^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2},\ s>a$
11.	$t^n e^{at},\ n>0$ integer	$\dfrac{n!}{(s-a)^{n+1}},\ s>a$
12.	$u_c(t)$	$\dfrac{e^{-cs}}{s}, s>0$
13.	$u_c(t)f(t-c)$	$e^{-cs}F(s)$
14.	$e^{ct}f(t)$	$F(s-c)$
15.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right),\ c>0$
16.	$\int_0^t f(t-\tau)g(\tau)d\tau$	$F(s)G(s)$
17.	$\delta(t-c)$	e^{-cs}

Laplace Transform Table

Table for Elementary Laplace Transforms

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
18.	$f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - \cdots - f^{(n-1)}(0)$

Definition 11.37

If $f(t)$ and $g(t)$ are defined on $[0, \infty)$, then the convolution $f * g$ is the function defined by

$$(f * g)(t) = \int_0^t f(t-\tau) g(\tau) d\tau = \int_0^t f(\tau) g(t-\tau) d\tau. \quad \square \tag{43}$$

Theorem 11.38

If $F(s) = \mathcal{L}\{f(t)\}$ and $G(s) = \mathcal{L}\{g(t)\}$ both exist for $s > a \geq 0$, then

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s), \ s > a. \quad \square \tag{44}$$

Shifting theorems and Step functions

Example 11.39

Find the solution of the differential equation

$$2y'' + y' + 2y = g(t), \quad y(0) = 0, \quad y'(0) = 0. \tag{45}$$

where

$$g(t) = u_5(t) - u_{20}(t) = \begin{cases} 1, & 5 \le t < 20, \\ 0, & 0 \le t < 5 \quad \text{and} \quad t \ge 20. \end{cases} \tag{46}$$

The Laplace transform of Eq.(45) is

$$\begin{aligned} 2s^2 Y(s) - 2sy(0) - 2y'(0) + sY(s) - y(0) + 2Y(s) &= \mathcal{L}\{u_5(t)\} - \mathcal{L}\{u_{20}(t)\} \\ &= (e^{-5s} - e^{-20s})/s. \end{aligned}$$

Using the initial values and solving for $Y(s)$, we obtain

$$Y(s) = \frac{e^{-5s} - e^{-20s}}{s(2s^2 + s + 2)}. \tag{47}$$

Example continue

To find $y = \phi(t)$, it is convenient to write $Y(s)$ as

$$Y(s) = (e^{-5s} - e^{-20s})H(s), \quad H(s) = \frac{1}{s(2s^2 + s + 2)}. \tag{48}$$

Then, if $h(t) = \mathcal{L}^{-1}\{H(s)\}$, we have

$$y = \phi(t) = u_5(t)h(t-5) - u_{20}(t)h(t-20). \tag{49}$$

Finally, to determine $h(t)$, we use the partial fraction expansion of $H(s)$:

$$H(s) = \frac{a}{s} + \frac{bs + c}{2s^2 + s + 2}. \tag{50}$$

Upon determining the coefficients, we find that $a = 1/2$, $b = -1$, and $c = -1/2$. Thus

$$\begin{aligned} H(s) &= \frac{1/2}{s} - \frac{s + 1/2}{2s^2 + s + 2} = \frac{1/2}{s} - \frac{1}{2}\frac{(s + 1/4) + 1/4}{(s + \frac{1}{4})^2 + \frac{15}{16}} \\ &= \frac{1/2}{s} - \frac{1}{2}\Big[\frac{s + 1/4}{(s + \frac{1}{4})^2 + (\frac{\sqrt{15}}{4})^2} + \frac{1}{\sqrt{15}}\frac{\sqrt{15}/4}{(s + \frac{1}{4})^2 + (\frac{\sqrt{15}}{4})^2}\Big]. \end{aligned} \tag{51}$$

Example continue

By Theorem 11.33

$$\mathcal{L}^{-1}\Big\{\frac{s+1/4}{(s+\frac{1}{4})^2+(\frac{\sqrt{15}}{4})^2}\Big\} = e^{-t/4}\cos\frac{\sqrt{15}}{4}t, \tag{52}$$

since

$$\mathcal{L}^{-1}\Big\{F(s)=\frac{s}{(s)^2+(\frac{\sqrt{15}}{4})^2}\Big\} = \cos\frac{\sqrt{15}}{4}t, \quad F(s-(-1/4)) = \frac{s+1/4}{(s+\frac{1}{4})^2+(\frac{\sqrt{15}}{4})^2}$$

and

$$\mathcal{L}^{-1}\Big\{\frac{\sqrt{15}/4}{(s+\frac{1}{4})^2+(\frac{\sqrt{15}}{4})^2}\Big\} = e^{-t/4}\sin\frac{\sqrt{15}}{4}t, \tag{53}$$

Thus, we obtain

$$h(t) = \frac{1}{2} - \frac{1}{2}\left[e^{-t/4}\cos(\sqrt{15}t/4) + (\sqrt{15}/15)e^{-t/4}\sin(\sqrt{15}t/4)\right]. \tag{54}$$

Unit impulses and the Dirac delta functions

Example 11.40

Find the solution of the initial value problem

$$2y'' + y' + 2y = \delta(t-5), \tag{55}$$

$$y(0) = 0, \quad y'(0) = 0. \tag{56}$$

This initial value problem arises from the study of the electric circuit or mechanical oscillator (can be regarded as the damped oscillator subject to the unit impulse $\delta(t-5)$). The only difference is in the forcing term. To solve the given problem, we take the Laplace transform of the differential equation and use the initial conditions, obtaining

$$(2s^2 + s + 2)Y(s) = e^{-5s}.$$

Thus

$$Y(s) = \frac{e^{-5s}}{2s^2 + s + 2} = \frac{e^{-5s}}{2} \frac{1}{(s + \frac{1}{4})^2 + \frac{15}{16}}. \tag{57}$$

Unit impulses and the Dirac delta functions

Example continue

By Theorem 11.33

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+\frac{1}{4})^2+\frac{15}{16}}\right\} = \frac{4}{\sqrt{15}}e^{-t/4}\sin\frac{\sqrt{15}}{4}t. \tag{58}$$

Hence, by Theorem 11.35, we have

$$y = \mathcal{L}^{-1}\{Y(s)\} = \frac{2}{\sqrt{15}}u_5(t)e^{-(t-5)/4}\sin\frac{\sqrt{15}}{4}(t-5), \tag{59}$$

which is the formal solution of the given problem. It is also possible to write y in the form

$$y = \begin{cases} 0, & t < 5, \\ \frac{2}{\sqrt{15}}e^{-(t-5)/4}\sin\frac{\sqrt{15}}{4}(t-5). & t \geq 5, \end{cases} \tag{60}$$

The convolution theorem

Example 11.41

Find the solution of the initial value problem

$$y' + \int_0^t y(\tau)(t-\tau)d\tau = \delta(t-1), \tag{61}$$

$$y(0) = 0. \tag{62}$$

Note that the integral is a **convolution**:

$$\int_0^t y(\tau)(t-\tau)\,d\tau = (y * t)(t)$$

That is, convolution of $y(t)$ with $f(t) = t$. Recall:

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

The convolution theorem

Example continue

Apply Laplace Transform:

- $\mathcal{L}\{y'\} = sY(s) - y(0) = sY(s)$ (since $y(0) = 0$)
- $\mathcal{L}\left\{\int_0^t y(\tau)(t-\tau)\,d\tau\right\} = Y(s) \cdot \frac{1}{s^2}$
- $\mathcal{L}\{\delta(t-1)\} = e^{-s}$

$$sY(s) + Y(s) \cdot \frac{1}{s^2} = e^{-s}$$

$$Y(s) = e^{-s} \cdot \frac{s^2}{s^3+1} = e^{-s}\left(\frac{1/3}{s+1} + \frac{\frac{2}{3}s - \frac{1}{3}}{s^2-s+1}\right) = e^{-s}\left(\frac{1}{3} \cdot \frac{1}{s+1} + \frac{1}{3} \cdot \frac{2s-1}{s^2-s+1}\right)$$

Define:

$$F(s) = \frac{1}{3} \cdot \frac{1}{s+1} + \frac{1}{3} \cdot \frac{2s-1}{s^2-s+1}, \quad f(t) = \mathcal{L}^{-1}\left\{\frac{1}{3} \cdot \frac{1}{s+1} + \frac{1}{3} \cdot \frac{2s-1}{s^2-s+1}\right\}$$

Use time-shifting property: $\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{e^{-cs}F(s)\} = f(t-1)u_c(t)$ (here $c = 1$)

The convolution theorem

Example continue

$$\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t} \Rightarrow \frac{1}{3}e^{-t}$$

$$\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2-s+1}\right\} = \mathcal{L}^{-1}\left(\frac{2\left(s-\frac{1}{2}\right)}{\left(s-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right)$$

$$= 2e^{t/2}\mathcal{L}^{-1}\left(\frac{s}{s^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right) = 2e^{t/2}\cos\left(\frac{\sqrt{3}}{2}t\right)$$

So:

$$f(t) = \frac{1}{3}e^{-t} + \frac{1}{3}\cdot 2e^{t/2}\cos\left(\frac{\sqrt{3}}{2}t\right) = \frac{1}{3}e^{-t} + \frac{2}{3}e^{t/2}\cos\left(\frac{\sqrt{3}}{2}t\right)$$

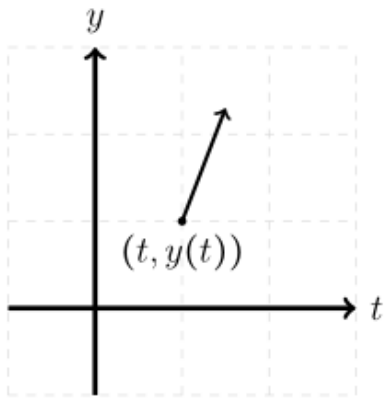
$$y(t) = f(t-1)u_1(t) = \left[\frac{1}{3}e^{-(t-1)} + \frac{2}{3}e^{(t-1)/2}\cos\left(\frac{\sqrt{3}}{2}(t-1)\right)\right]u_1(t)$$

First-order equation

Given a first order nonlinear ODE

$$y'(t) = f(t, y),$$

for continuous functions f and $\frac{\partial f}{\partial y}$, we know that there is exactly one solution to the IVP when initial conditions are given. Furthermore we can plot the graph of t vs y using the equation.



At each point (t, y) we can draw a line segment with the slope $f(t, y)$. This gives a **direction field** for the ODE. Usually, one can also put an arrow at the end of each line segment (the arrow pointing out the evolution direction of the solution when increasing t), this is called **direction field** or **slope field**.

Trajectory for ODE

If we put a particle inside this direction field at initial position (t_0, y_0), the particle will move along with the direction field, this traces out a **trajectory** $\{(t, y(t)) : t \in I\}$ for the solution to the ODE.

Definition of stability

For the stationary point y_*, if we start close to y_*, and the trajectory moves towards y_* as time progresses, then we call this the stationary point y_* is **asymptotically stable**.

Definition 11.42

(Stability). Given an autonomous first order ODE $y' = f(y)$, and a stationary solution y_*. We say that y_* is **asymptotically stable** if there is an $\delta_0 > 0$ (depending only on y_*) such that for any solution $\phi(t)$ to the IVP

$$y' = f(y) \text{ for } t \in I, \quad y(t_0) = y_0 \text{ with } t_0 \in I,$$

the following property is satisfied:

$$|y_0 - y_*| < \delta_0 \quad \Longrightarrow \quad \phi(t) \to y_* \text{ as } t \to \infty,$$

which means if we start close to y_*, we will move towards y_* as time progresses.

Definition of stability

For the stationary point y_*, if we start close to y_*, the trajectory moves far away from y_* as time progresses, then we call this the stationary point is **unstable**.

Denote $\phi(t)$ be the solution to the IVP

$$y' = f(y) \text{ for } t \in I, \quad y(t_0) = y_0 \text{ with } t_0 \in I,$$

Then for an unstable stationary solution y_*, except for $y_0 = y_*$(which implies that $\phi(t) = y_*$ for all $t \geq t_0$), any other initial condition would lead to $|\phi(t) - y_*| \nrightarrow 0$ $(t \to +\infty)$, and so the solution $\phi(t)$ will never reach y_* for an unstable stationary solution $y = y_*$.

Critical point and stationary solutions

Definition 11.43

(Critical point and stationary solutions). Given a continuous function $f(t, y)$, suppose $y_* \in \mathbb{R}$ is a point such that

$$\boxed{f(t, y_*) = 0 \quad \forall t \in I},$$

then y_* is a **critical point** of f. We call the constant function

$$\phi(t) = y_* \quad \forall t \in I$$

a **stationary solution** to the ODE $y' = f(t, y)$.

Autonomous ODE

Besides the direction fields, another useful graphic for the nonlinear **autonomous** ODE

$$y' = f(y)$$

is the graph y vs $f(y)$.

Example 11.44

Recall the Logistic equation: for positive constants r and K,

$$y' = f(y) = ry\left(1 - \frac{y}{K}\right)$$

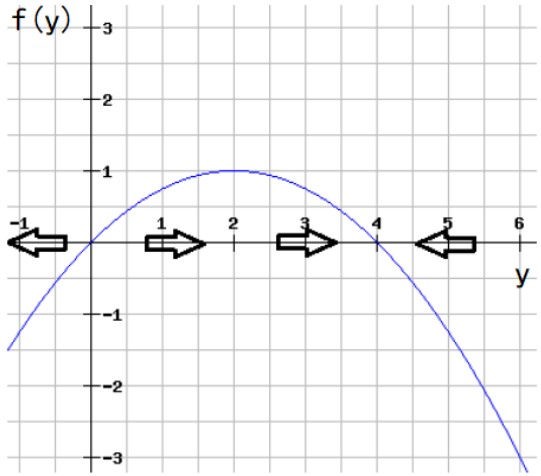
for population dynamics. It is easy to see that $y = 0$ and $y = K$ are stationary solutions, and if y is not equal to 0 or K, then

$$y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}} \quad \textit{for } y(0) = y_0.$$

Hence, we can deduce that for nonnegative initial values y_0,

$$y(t) \to K \textit{ as } t \to \infty \textit{ if } y_0 > 0, \quad y(t) = 0 \textit{ for all } t > 0 \textit{ if } y_0 = 0.$$

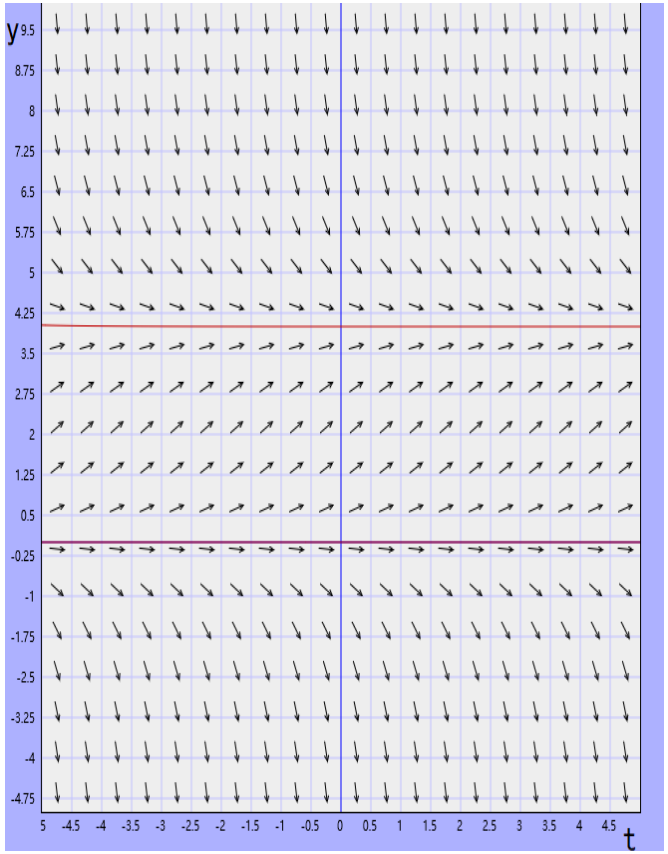
Logistic equation



The plot y vs $f(y)$ for the Logistic equation. Arrows indicate the behaviour of the solution.

Logistic equation

We can plot the direction field for the case $r = 1$ and $K = 4$ in the following figure, and observe that the line segments with y-coordinate equal to 0 or 4 ($y = 0$ or $y = 4$) have zero slope.



Direction field for the Logistic equation in (t, y) plane with $r = 1$ and $K = 4$ with arrows indicating the changing of y.